Andrew Park

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Research Interest

My research interests lie in **Computer Architecture** and **Compilers**, especially for **GPU**s. I'm especially interested in making GPU programming easier by better architectures and compilers.

EDUCATION

Purdue University, West Lafayette, IN **August 2025 -**
Ph.D., Electrical and Computer Engineering

Yonsei University, Seoul, Republic of Korea **August 2025**
BS, Electrical and Electronic Engineering
Thesis: "Limitation of Optimization Targets on Loop Optimizers"

- Advisor: Prof. Hanjun Kim
- Relevant Coursework: Multicore and GPU programming, Computer Architecture, Microprocessors, System IC Design

RESEARCH EXPERIENCE

Yonsei University, Seoul, Republic of Korea **March – June 2024**
Undergraduate Thesis Student
Advisor: Prof. Hanjun Kim
Thesis: "Limitation of Optimization Targets on Loop Optimizers"

- Identified LLVM's polyhedral optimizer Polly's limitations in optimizing codes with dynamically bounded loops
- Restructured Polly's target-setting modules to uncover all loops accurately for 3.5x faster optimization results

Seoul National University, Seoul, Republic of Korea **June – August 2024**
Undergraduate Research Assistant
Project: Development of Near-Data computing solution targeting MoE LLMs.
Advisor: Prof. Hyunkjae Lee

- Studied mixture-of-experts models' functionality and characteristics
- Implemented a near-data computing solution that alleviates bottlenecks between system memory and device using Verilog on AMD/Xilinx U280 FPGA
- Achieved 2.7x gain for 'google/switch-base-8' summarization task

PROJECTS

IEEE754 single precision FPU development **May – August 2024**

- Studied various types of adders; applied Kogge-Stone adder
- Implemented all four operations using Verilog
- Implemented adder multiplexing scheme to optimize resource utilization based on observed patterns of adder usage while FPU is performing its functions
- Implemented control circuit to monitor calculation status and generate quick finish signal

WORK EXPERIENCE

Insight Edu Inc., Seoul, Republic of Korea **Jan 2019 - Nov 2021, Mar 2024 - Present**
Developing team leader for development of learning management system,
developed a Learning Management System.

Seoul National School for the Blind, Seoul, Republic of Korea **Jan 2022 - Oct 2023**
Assistant teacher for visually impaired students

TEACHING EXPERIENCE

Advanced C Programming (ECE264) **Fall 2025**
Graduate Teaching Assistant

AWARDS

Merit-based scholarship, Yonsei University College of Engineering **Fall 2024**
Merit-based scholarship, Yonsei University College of Engineering **Spring, 2024**
Educational Outreach Scholarship, Yonsei University **2019**
Life Academy Scholarship, Yonsei University **2019**

SKILLS AND LANGUAGES

Languages: C, C++, Verilog, MATLAB, Python, Javascript, React
Technical: LLVM, Polly, GCC, Xcelium, Vivado, Vitis, v++, Synopsys Design Compiler, Quartus, Pybind, AWS, GCP, Firebase
General: English (Fluent), Korean (Native)